AWS Machine Learning

Cloud Computing

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Contents

Learning Outcomes

- Overview of Machine Learning
- Intro to AWS Machine Learning Pipelines

Reading

- https://aws.amazon.com/ai/machine-learning/
- https://aws.amazon.com/sagemaker/ai/
- https://docs.aws.amazon.com/sagemaker/
- https://github.com/aws/amazon-sagemaker-examples/tree/main

What is Machine Learning (ML)

A subset of Artificial Intelligence (AI) for programming computers to learn prediction rules from data.

Machine Learning is well suited for:

- Problems for which existing solutions require a lot of fine-tuning or long lists of rules.
- **Complex problems** for which using a traditional approach yields no good solution.
- **Fluctuating environments**: a Machine Learning system can adapt to new data.
- **Getting insights** about complex problems and large amounts of data.

Terminology

Artificial intelligence - Machines performing human tasks

Machine learning - Training models to make predictions

Deep learning - Transformer-based neural networks

Large language models (LLMs) - General-purpose (aka foundational) models pre-trained on vast amounts of data & suitable for a variety of tasks without further training.

Generative AI - Type of AI that can create new content

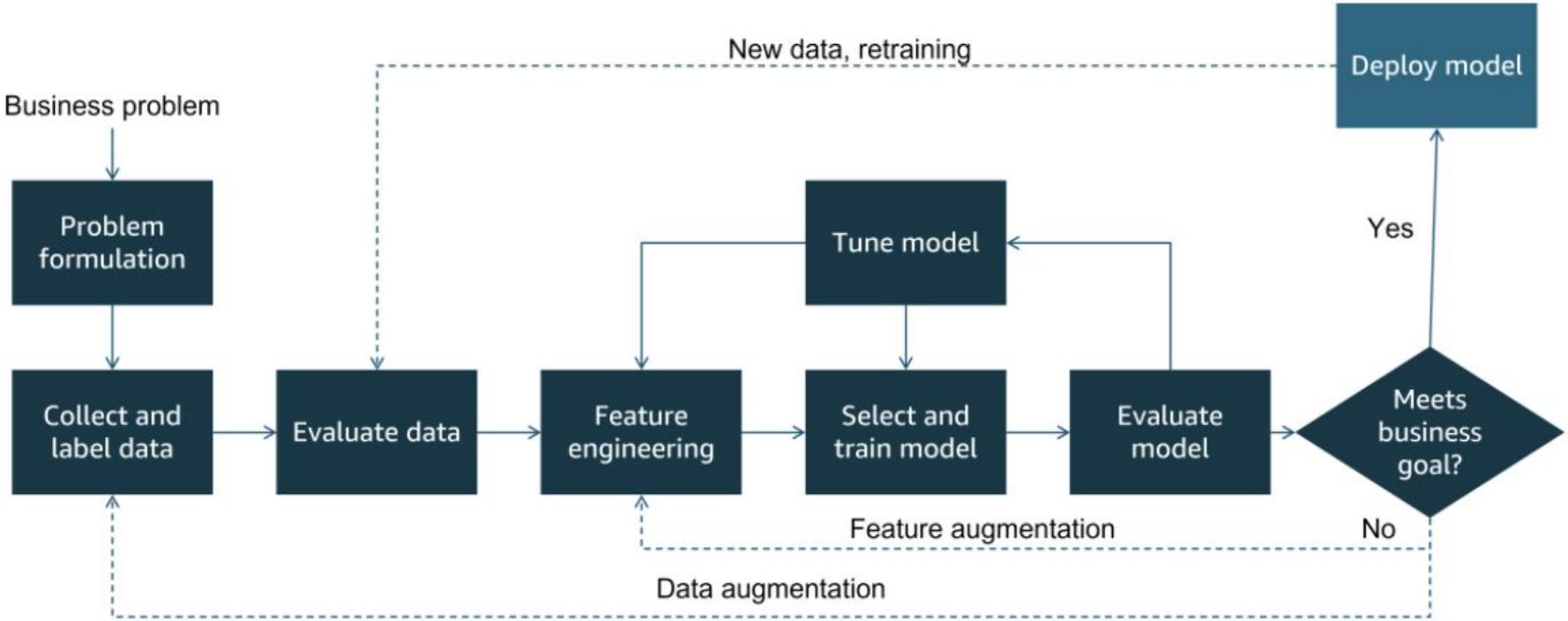
Why Cloud ML?

- Data preparation can involve large data sets or data in production systems
- ML is compute & memory intensive
- Parallelize ML operations through distributed computing
- Managed infrastructure allows user to focus on ML problems
- Cloud providers support encryption, security, governance, and auditing
- To integrate models with online applications for real-time inference

Machine Learning Process

- Problem formulation
- Data acquisition & cleaning
- Model training
  - Feature engineering - process of selecting or creating the features used to train a model
  - Training
  - Evaluating - measuring how well the model performs against new data
  - Tuning - modifying a model's input data, features, or hyperparameters & re-trainng until finding an model with the best results
- Deployment

ML pipeline:



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Storing data in AWS

Amazon Simple Storage Service (Amazon S3)

Amazon FSx

Amazon Elastic File System (Amazon EFS)

Amazon Relational Database Service (Amazon RDS)

Amazon Redshift

Amazon Timestream

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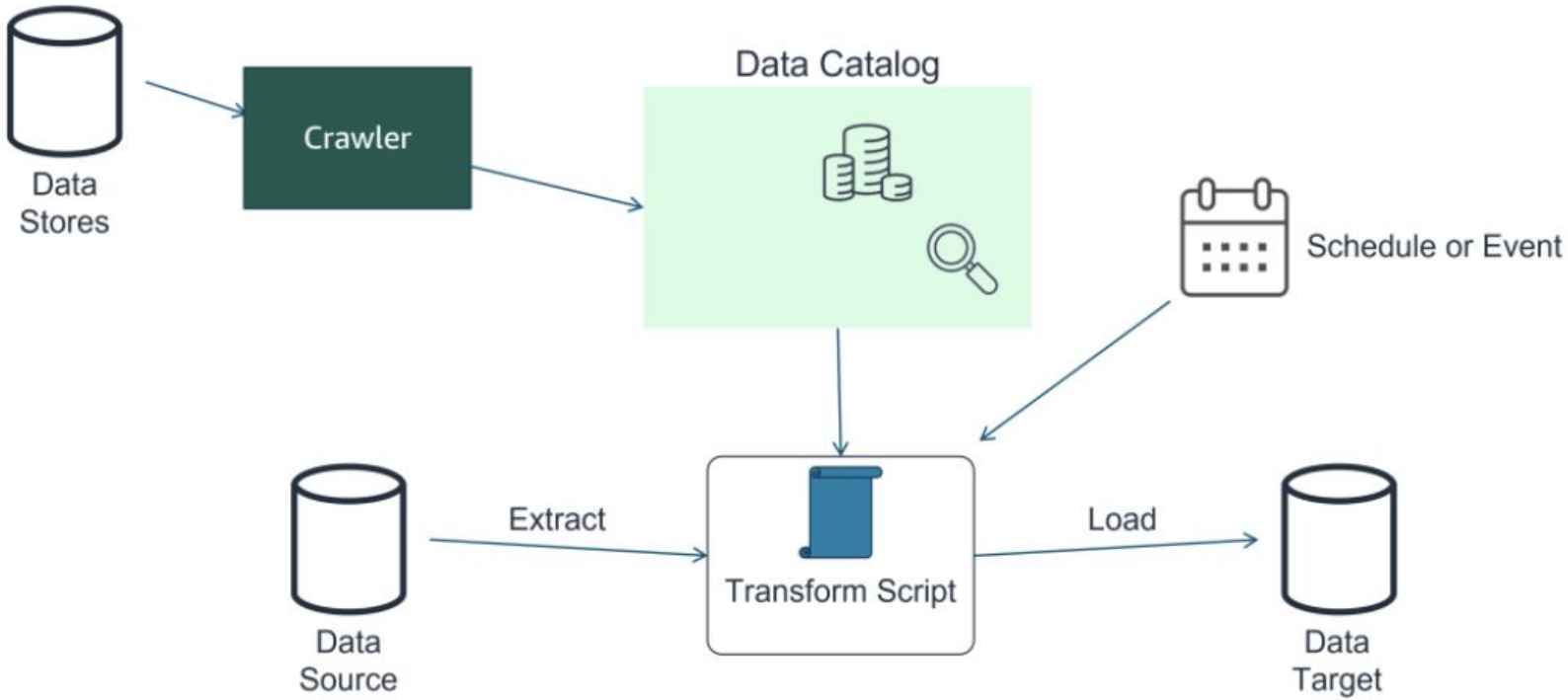
AWS Storage Services

- **AWS S3** - optimal for reliable file storage. Can trigger a Lambda function to run AWS Glue ETL jobs when new data arrives
- **Amazon FSx for Lustre** - serves S3 files to SageMaker at high speeds for training multiple models from same datasets
- **Amazon EFS** - can directly launch model training jobs from the service without the need for data movement.

Extract, transform, load (ETL)



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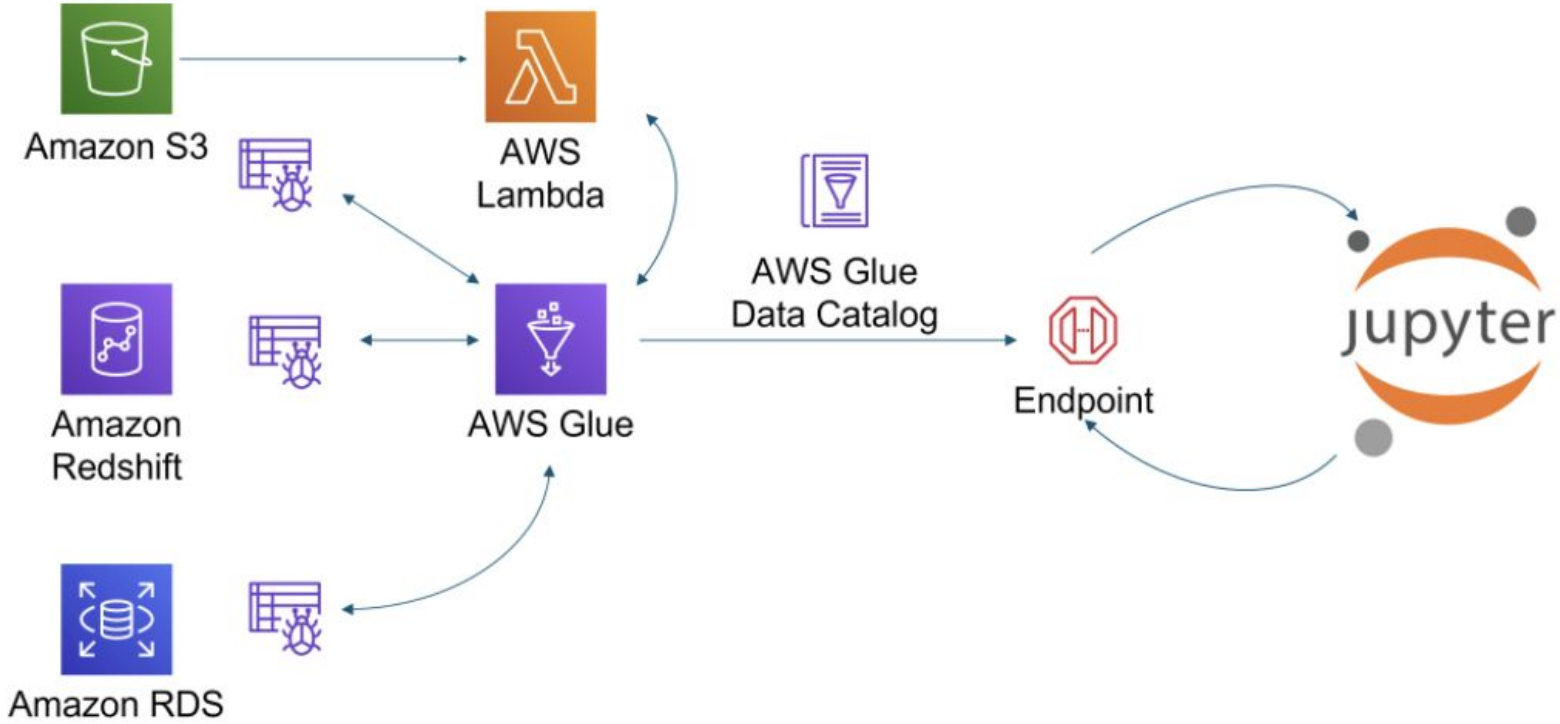
AWS Glue

- A central metadata repository - the AWS Glue Data Catalog
- An ETL engine that automatically generates Python or Scala code
- A flexible scheduler that handles dependency resolution, job monitoring, and retries
- Works with semistructured data using **dynamic frames**, similar to Apache Spark DataFrame but self-describing (no schema).
- Has special advanced transformations designed for dynamic frames
- **AWS Glue** console can discover data, transform it, and make it available for search and querying. Calls the underlying services to orchestrate data transformations.

AWS Glue overview



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Amazon SageMaker

Cloud-based system allowing users to build, train, customize, and deploy AI models at scale. Provides:

SageMaker Studio - integrated development environment supporting the full ML life cycle

- Can launch fully managed JupyterLab or RStudio
- Code editor
- Visual Studio code integration
- Supports AWS Glue interactive sessions

https://aws.amazon.com/sagemaker/ai/studio/

Amazon SageMaker

Ground Truth

Set up and manage labeling jobs for highly accurate training datasets by using active learning and human labeling.

Notebook

Provide AWS and SageMaker SDKs and sample notebooks to create training jobs and deploy models.

Training

Train and tune models at any scale. Use high-performance AWS algorithms, or bring your own.

Inference

Create models from training jobs, or import external models for hosting so you can run inferences on new data.

AWS Marketplace

Find, buy, and deploy ready-to-use model packages, algorithms, and data products in AWS Marketplace.

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SageMaker - Data Preparation

SageMaker offers several approaches to inspect, clean, & transform data

- **Data Wrangler** (in SageMaker **Canvas**) - a visual low-code tool allowing users to combine, transform, and clean datasets through point-and-click interactions.
- **SageMaker Studio SQL** extension - allows users to connect to AWS storage services, author ad-hoc SQL queries, and operate on the results in **JupyterLab** notebooks with python scripts
- **Elastic Map Reduce (EMR)** serverless - directly access EMR applications from Studio notebooks for large-scale data preparation using open-source frameworks such as Apache Spark and Apache Hive.

SageMaker Notebook Instances

- Useful for orchestrating ML pipeline operations
- Can use **SageMaker Python SDK**, which abstracts the AWS SDK for Python
- Supports a range of instance types optimized for ML operations

https://docs.aws.amazon.com/sagemaker/latest/dg/nbi.html

Common ML Tools

Python

- Jupyter notebook
- Jupyter Lab
- Python libraries
  - pandas
  - matplotlib
  - Seaborn
  - NumPy
  - Scikit-learn
- Frameworks
  -

Machine learning frameworks and infrastructure

Machine learning **frameworks** provide tools and code libraries:

- Customized scripting
- Integration with AWS services
- Community of developers

Amazon **instances** that are designed for machine learning applications:

- AWS IoT Greengrass provides an infrastructure for building machine learning for IoT devices
- Amazon Elastic Inference reduces costs for running machine learning applications

PyTorch	Caffe2	Torch
TensorFlow	Gluon	Chainer
Keras	CNTK	Apache MXNet

EC2 P3 instances

EC2 C5 and C5n instances

AWS IoT Greengrass

Amazon Elastic Inference

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Model Training

- SageMaker supports common file formats
- Most Amazon SageMaker algorithms work best when you use the optimized **protobuf recordIO** format for the training data. This format allows SageMaker to stream data from S3 in **Pipe** mode for faster start times for training jobs and better throughput.
- Training options
  - built-in algorithms
  - supported frameworks - prebuilt containers to support deep learning frameworks
  - prebuilt Docker images for common ML libraries
  - custom frameworks -
  - AWS Marketplace - ready-to-use algorithms & models
-
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Amazon SageMaker built-in algorithms

Classification	Quantitative	Recommendations	Unsupervised	Specialized	
XGBoost	XGBoost	Factorization machines	K-means	Image classification	Neural Topic Model (NTM)
Linear learner	Linear learner		Principal Component Analysis	Sequence-to-sequence	Latent Dirichlet Allocation (LDA)

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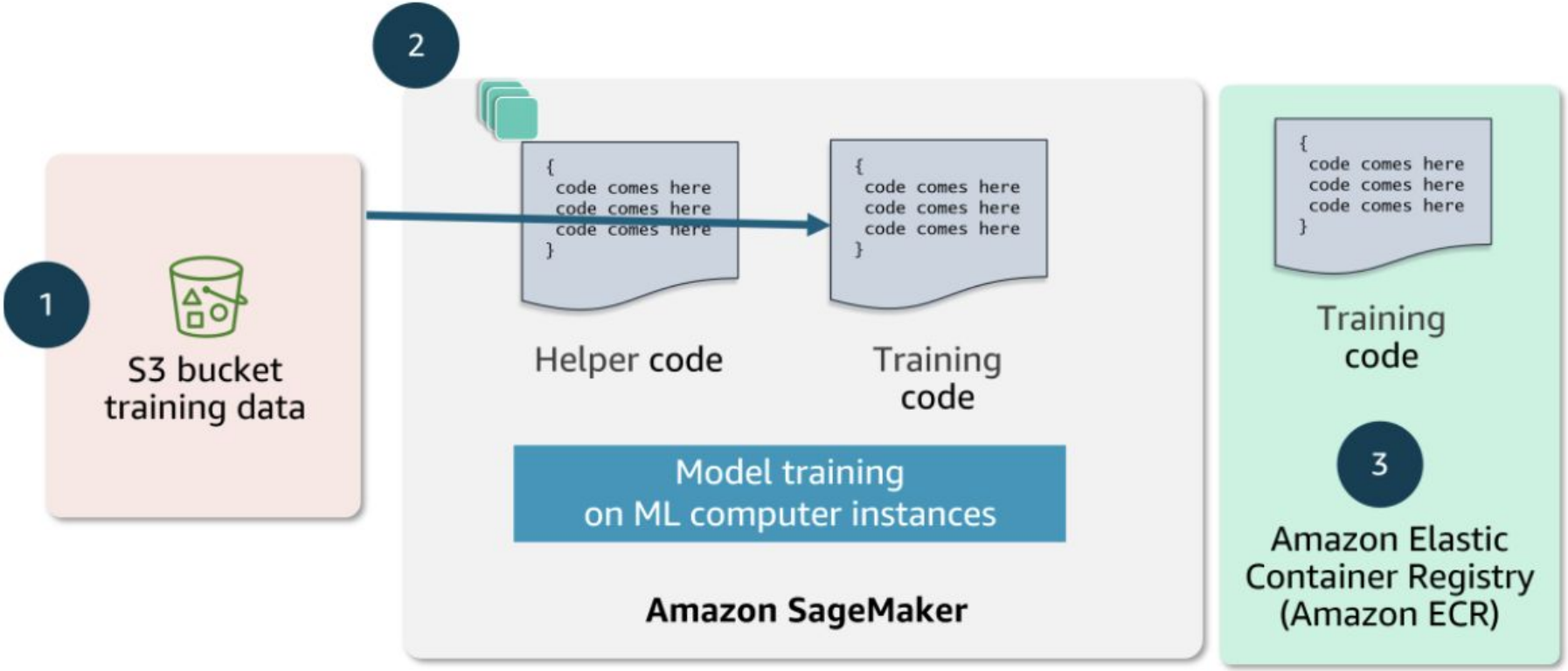
Creating a training job in Amazon SageMaker



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Model Training

- Train model with a training job that includes
  - S3 url for input data
  - ML compute instance (type & size) to use
  - S3 url for output data
  - Amazon Elastic Container Registry (ECR) where training code is stored